

BLOWFISH

AIM

To implement the **Blowfish symmetric encryption algorithm** for encrypting and decrypting a given plaintext using a secret key.

ALGORITHM

1. Start the program.
2. Read the plaintext and secret key from the user.
3. Create a Blowfish cipher using the given key.
4. Encrypt the plaintext and display the ciphertext.
5. Initialize the cipher in decryption mode using the same key.
6. Decrypt the ciphertext.
7. Display the original plaintext.
8. Stop the program.

PROGRAM

```
import javax.crypto.Cipher;
import javax.crypto.spec.SecretKeySpec;
import java.util.Base64;
import java.util.Scanner;

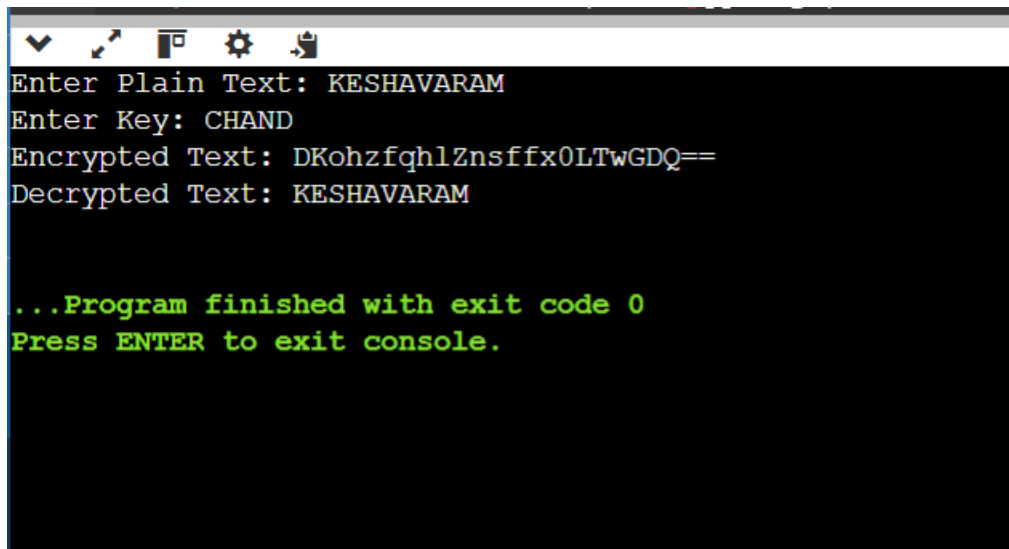
public class Main {
    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Plain Text: ");
        String text = sc.nextLine();

        System.out.print("Enter Key: ");
        String key = sc.nextLine();
```

```
SecretKeySpec secretKey =  
    new SecretKeySpec(key.getBytes(), "Blowfish");  
  
Cipher cipher = Cipher.getInstance("Blowfish");  
  
cipher.init(Cipher.ENCRYPT_MODE, secretKey);  
byte[] encrypted = cipher.doFinal(text.getBytes());  
  
System.out.println("Encrypted Text: " +  
    Base64.getEncoder().encodeToString(encrypted));  
  
cipher.init(Cipher.DECRYPT_MODE, secretKey);  
byte[] decrypted = cipher.doFinal(encrypted);  
  
System.out.println("Decrypted Text: " +  
    new String(decrypted));  
  
sc.close();  
}  
}
```

A screenshot of a terminal window with a dark background. The window has a title bar with standard OS icons (minimize, maximize, close) and a toolbar with icons for a dropdown menu, a window, a gear, and a document. The text in the terminal is as follows:

```
Enter Plain Text: KESHAVARAM
Enter Key: CHAND
Encrypted Text: DKohzfqhlZnsffx0LTwGDQ==
Decrypted Text: KESHAVARAM

...Program finished with exit code 0
Press ENTER to exit console.
```

RESULT

Thus, the **Blowfish encryption and decryption algorithm** was implemented successfully, and the original plaintext was recovered after decryption.

